

We established that :

$$\partial^{\mathbf{n}}(\delta\Pi - d\Gamma_x^*\Pi_x)(x) = 0 \quad \text{for } |\mathbf{n}| < 2 + s. \quad (1)$$

We will see how this specifies uniquely the coefficients $d\Gamma_x^*$. Indeed, by definition,

$$d\Gamma_x^* := \sum_{|\mathbf{n}| < 2+s} d\pi_x^{(\mathbf{n})} \Gamma_x^* \partial_{z_{\mathbf{n}}}.$$

And as

$$\partial_{z_{\mathbf{n}}} z_{\mathbf{m}} = \delta_{\mathbf{n}, \mathbf{m}},$$

It follows that

$$\forall \mathbf{m}, |\mathbf{m}| \geq 2 + s, d\Gamma_x^* z_{\mathbf{m}} = 0.$$

Additionnaly, $d\Gamma_x^*$ behaves like a derivation :

$$\begin{cases} d\Gamma_x^* \pi \pi' = (d\Gamma_x^* \pi)(\Gamma_x^* \pi') + (\Gamma_x^* \pi)(d\Gamma_x^* \pi'), \\ d\Gamma_x^* 1 = 0, \\ d\Gamma_x^* z_3 = 0. \end{cases}$$

This follows directly from the behaviour of $\partial_{z_{\mathbf{n}}}$ which is a derivation, and Γ_x^* which is an algebra morphism. As a consequence, together with its finiteness property, $d\Gamma_x^*$ is determined by its values on the finitely many coordinates $\{z_{\mathbf{n}}\}_{|\mathbf{n}| < 2+s}$.

Knowing $\delta\Pi$ and (1), we actually know $d\Gamma_x^* \partial^{\mathbf{n}} \Pi_x(x)$ for all $\mathbf{n}$ such that $|\mathbf{n}| < 2 + s$. We also know this fact about the jet:

$$\frac{1}{\mathbf{n}!} \partial^{\mathbf{n}} \Pi_x(x) - z_{\mathbf{n}} \in \mathbb{R}[[z_3, \{z_{\mathbf{m}}\}_{|\mathbf{m}| < |\mathbf{n}|}]]. \quad (2)$$

Now, an induction argument shows that we thus know $d\Gamma_x^* z_{\mathbf{n}}$ for all $\mathbf{n}$ such that $|\mathbf{n}| < 2 + s$.

Proof. The base case is clear because $\Pi_x(x) - z_0 \in \mathbb{R}[[z_3]]$ and $d\Gamma_x^* z_3 = 0$ so

$$d\Gamma_x^* z_0 = d\Gamma_x^* \Pi_x(x).$$

Now let $k \in \mathbb{N}$, $k < 2 + s$ be such that we know $d\Gamma_x^* z_{\mathbf{m}}$ for all $\mathbf{m}$ such that $|\mathbf{m}| < k$. For all $\mathbf{n}$ with $|\mathbf{n}| = k$, (2) shows that $d\Gamma_x^* z_{\mathbf{n}}$ is a function of $d\Gamma_x^* \partial^{\mathbf{n}} \Pi_x(x)$ and some $d\Gamma_x^* z_{\mathbf{m}}$ with $|\mathbf{m}| < k$, which are known by hypothesis. □

This will be useful to explicit a robust relation $\delta\Pi \longrightarrow \delta\Pi^-$. By robust we mean that the map doesn't rely on diverging constant such as c , like if we used the first definition of $\delta\Pi^-$:

$$\delta\Pi^- := (3z_3\Pi^2 + c)\delta\Pi + \delta\xi_{\rho}\mathbf{1}.$$

We claim that :

$$(\delta\Pi^- - \delta\xi_{\rho}\mathbf{1} - d\Gamma_x^* \Pi_x^-)(x) = 0 \quad (3)$$

Proof. Let's work out $d\Gamma_x^* \Pi_x^-(x)$:

$$\begin{aligned} d\Gamma_x^* \Pi_x^-(x) &= d\Gamma_x^* (z_3 \Pi_x^3 + c \Pi_x + \xi_{\rho} \mathbf{1})(x) \\ &= ((\Gamma_x^* z_3)(d\Gamma_x^* \Pi_x^3) + (\Gamma_x^* c)(d\Gamma_x^* \Pi_x) + d\Gamma_x^* \xi_{\rho} \mathbf{1})(x) \quad (\text{derivation properties}) \\ &= (3z_3(d\Gamma_x^* \Pi_x)(\Gamma_x^* \Pi_x^2) + c d\Gamma_x^* \Pi_x)(x) \quad (\text{same properties, and } \Gamma_x^* c = c, \Gamma_x^* z_3 = z_3) \\ &= (3z_3 \Pi^2 + c)(d\Gamma_x^* \Pi_x)(x) \quad (\text{because } \Gamma_x^* \Pi_x = \Pi) \\ &= (3z_3 \Pi^2 + c)(\delta\Pi)(x) \quad (\text{by (2) for } \mathbf{n} = \mathbf{0}) \\ &= (\delta\Pi^- - \delta\xi_{\rho})(x) \quad (\text{by definition of } \delta\Pi^-) \end{aligned}$$

□

The calculation is easier in this direction although we are truly going in the opposite direction. The success of our approach may be due to the fact that we use an approximation of $\delta\Pi$ at each x with the germ $d\Gamma_x^*\Pi_x$, as opposed to only utilizing $\Pi_0; \Pi_0^-$.